Harshit Shah *Systems Eng.*

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Magdeburg

Technical skills

CAD design & simulation

- Altair HyperWorks
- Fusion 360
- CATIA V5
- Synera

Matlab & Simulink

- Optimization Toolbox
- Simscape Battery
- Powertrain Blockset
- Global Optimization Toolbox

Courses

Design of control systems using MATLAB and Simulink, *Mathworks*

- Developed control loops using MATLAB and Simulink.
- Implemented PID and lead-lag controllers.
- Applied system identification for model estimation.
- Performed frequency domain analysis and linearization.

Education

MSc Systems Engineering for Manufacturing, *Otto Von Guericke University*
10/2022 – Present | Magdeburg, Germany

- Systems engineering for manufacturing systems
- Factory automation and industrial robotics
- Advanced applications of Industry 4.0
- Modeling and simulation of mechatronic systems

Profile

MSc student Systems Engineering (Manufacturing) with CAD/FEM experience and a focus on digital engineering & design automation. Confident in standard-compliant documentation, technical analysis and configuration-oriented work ; specifically targeting CATIA V5/DMU and KBE/Python in a targeted manner to reduce manual design effort and accelerate architecture/integration processes.

Professional experience

Master Thesis, *Otto Von Guericke University*
02/2026 – Present

Topic: CAD-based design automation for battery housings: automatic rough design based on variants and requirements with simulation/load cases and DMU review capability.

- State of the art in design automation, parametric CAD, and DMU (virtual reviews/variants) developed.
- Existing variant tool analyzed: results presentation checked, relevant geometric parameters identified.
- Requirements engineering: Requirements for (partially) automated rough design of the battery housing (installation space/packaging) defined.
- Models & simulation integrated: Load cases, crash, cell/module fixation, and PDU/connector/thermal management taken into account.
- CAD framework developed & verified; outputs prepared for DMU/ product structure navigation and reviews.

Volunteer, *ELARA Aerospace*
08/2024 – 04/2025 | Munich

- Creation of logic and simulation in Synera Workspace for cooling channels for the rocket engine
- Creation of topologies for the design space & Structural simulation of fuel tanks

Internship – Rail vehicle design & simulation, *DLR Institute of Vehicle Concepts*
04/2024 – 07/2024 | Stuttgart

- Design of a temporary underframe for inspecting Fr8Rail bogies using load case calculations and FEM simulations.
- Management of CAD design of a night train car body in CATIA V5, taking into account EBO (Gauss 2) and EN standards (EN 12663, EN 15227).
- Use of parametric design for modular vehicle development with a focus on coupling systems, track gauge flexibility, and load variants.

Academic projects

Altair Global Student Competition
03/2023 – 04/2023

- Performed topology optimization of the TL-403 pull-down latch to increase structural efficiency and reduce weight.
- Award for the best promotional video in May, emphasizing creativity, target audience appeal, and visual implementation.

Automation of Battery Pack Design, *Simulation and Modeling with MATLAB/Simulink*

- Automation of design using Simscape Battery.
- Simulation of electrical and thermal behavior.
- Optimization of cell configuration (series/parallel).
- Model generation via the Battery Builder app.
- BMS logic (SoC/SoH) in Stateflow.